

A Comparative Study of Retrieval-Augmented Generation (RAG) Chatbots

1st Kalindi Vijesh Parekh
Department of Computer Science
San Jose State University
San Jose, USA
kalindivijesh.parekh@sjsu.edu

2nd Navrati Saxena
Department of Computer Science
San Jose State University
San Jose, USA
navrati.saxena@sjsu.edu

3rd Mohammad Adil Ansari
Bentonville, USA
adil.ansari08642@gmail.com

Abstract—The use of Retrieval-augmented generation (RAG) in chatbot platforms has transformed academic spaces by significantly improving information accessibility. RAG has become a viable approach to upgrading Large Language Models (LLMs) with external knowledge access in real time. With the growing availability of advanced LLMs such as GPT, DeepSeek, Claude, Gemini, and Llama, there is a growing need to compare RAG systems based on different LLMs. This study compares the responses of four different RAG chatbots using popular LLMs against a uniquely designed evaluation dataset. Specifically, the study compares the responses and performance of closed-source (GPT-4o and Claude) and open-source models (DeepSeek and Llama) against questions requiring inference from multiple scientific corpora with intricate content and structure. All RAG models in this research use a Chroma vector database to store embeddings. The retrieved documents and the query are provided as input prompts to the LLMs, thus allowing contextually grounded response construction. Each chatbot is evaluated based on ten complex research papers from various domains in computer science. The specifically designed evaluation dataset contains 75 questions derived from these research papers, with a wide variety of questions ranging from simple yes/no questions to questions requiring an understanding of multiple papers. The responses provided by each chatbot are measured quantitatively using standard measures, including Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), and Bidirectional Encoder Representations from Transformers (BERT) scores, to evaluate response quality comprehensively.

Index Terms—Retrieval Augmented Generation, Generative AI, Chatbot, Claude, GPT, Claude, DeepSeek, Llama

I. INTRODUCTION

While the use of Large Language Models (LLMs) has experienced unmatched utilization across different sectors of academia, they still suffer from inherent limitations. These limitations are significant because they can generate false information due to insufficient knowledge, provide outdated answers, utilize non-authoritative information sources, and produce errors based on vagueness in terminology. These problems occur because their pre-trained data limits LLMs, leading to unreliable outputs. Retrieval Augmented Generation (RAG) overcomes these drawbacks by incorporating real-time external knowledge into the generation process, making it more accurate and relevant. It combines the strength of retrieval systems for domain-specific information

and the generative abilities of LLMs to generate ground solutions [1] [2].

As more powerful and advanced LLM models become available for users, it is critical to evaluate how different models perform against each other when embedded into an RAG architecture. The primary objective of this research is to present a detailed comparative study of the responses provided by different RAG chatbots using four widely popular LLMs when questioned against a uniquely curated evaluation dataset generated from a pre-selected corpus of complex scientific literature. Understanding research papers with complex concepts is a significant obstacle for most students. LLMs cannot handle subtle questions, particularly those that require understanding from multiple such papers. RAG-based chatbots allow users to extract key points from multiple sources. Therefore, evaluating these systems against a corpus of sophisticated academic material provides valuable insights into understanding which model performs better than the other against standard retrieval mechanisms.

This study compares four RAG-based chatbots, each built on a different foundation model—two closed-source (GPT-4o, Claude) and two open-source (DeepSeek, Llama). These chatbots, all built on the LangChain framework with Chroma DB as the vector store, feature an intuitive user interface that allows users to easily upload research papers. For this research, we upload a selected corpus of ten arXiv research papers on diverse computer science subjects paired in groups of two so that each paper in the group is, in some way, interconnected with the other papers. The evaluation data, consisting of 75 questions (15 for each group), is specially designed to contain questions ranging from simple factual, yes/no, to summary and reasoning, which would require understanding of both the papers to answer the questions. The responses are then graded against reference answers by standard metrics such as BLEU-1, BLEU-4, ROUGE-1, ROUGE-2, and ROUGE-L. The result provides an overall comparison of answer correctness and adequacy among these models and the potential of RAG-enhanced LLMs in educational settings.

II. BACKGROUND AND RELATED WORKS

The application of RAG chatbots in academia has significantly helped across 3 key areas: data access, research assistance, and student advising. One of the most significant applications of RAG chatbots in academia is research assistance. A research paper introduced ChatPapers, an AI chatbot with a data source of around 200,000 research papers, and tested with 30 sample questions in computer science. Based on Readability Metrics, the LLM model produced better outcomes than standard keyword prompt answers. To enable effective literature discovery, the system uses Facebook AI Similarity Search (FAISS) indexing and semantic search embeddings [3]. Similarly, [4] highlighted how RAG-driven academic chatbots have an advantage in data retrieval due to being able to store better and reference past conversations. This is a great benefit for research as it allows the proper tracing of ideas and data to past papers.

RAG chatbots have also been beneficial in academic advising, as highlighted in [5], where the team developed an LLM-based chatbot that extracts student data from transcripts to provide targeted advice. Students were provided with quality advice on choosing courses, monitoring graduation requirements, and optimizing learning paths by having upfront access to previously hard-to-find information. Outside of research and counseling, [6] also demonstrates how RAG-based LLMs can be leveraged to provide accurate and timely information about queries regarding admissions, programs, faculties, and campus facilities. A study [7] introduced OpenRAG, an open-source retrieval-augmented generation framework for online learning platforms. This framework can generate customized learning paths based on individual students' needs and preferences, thus improving engagement and reducing comprehension times. Apart from student-facing applications, [8] demonstrated how RAG-powered chatbots can help overcome issues with prolonged usage of e-books. This PDF-driven chatbot enables students and staff to easily traverse PDFs for specific topics by simply asking the chatbot, thereby removing the need to read the entire document. Overall, the advent of RAG-based chatbots in the academic world indicates their increasing capability for automating academic support, aiding and enhancing research methodologies, and streamlining administrative work.

III. PROPOSED WORK

A. Architecture of RAG systems

The basic crux of our work involves designing intelligent chatbots using RAG from scratch. Fig. 1 shows the flow of our proposed system, highlighting the different phases involved in the process. Our RAG chatbot consists of three main phases.

1) *Data Ingestion and Processing Pipeline*: The user enters research articles through an easy-to-use interface, which the system uses as its base corpus. The input documents are then broken into chunks of standard size (1024 tokens here) to allow them to be quickly processed. The chunks are passed

through the embedding language models, transforming the text into a numeric representation. The output embeddings thus obtained are then stored in a vector database, allowing quick query and retrieval. We use ChromaDB [9], a publicly available vector database tailored to manage and query vector embeddings. This database serves as the knowledge base, supporting the technique of semantic vector searches—how the system identifies data points semantically similar to the questions being asked. The system uses this process to ensure the information provided is accurate and relevant.

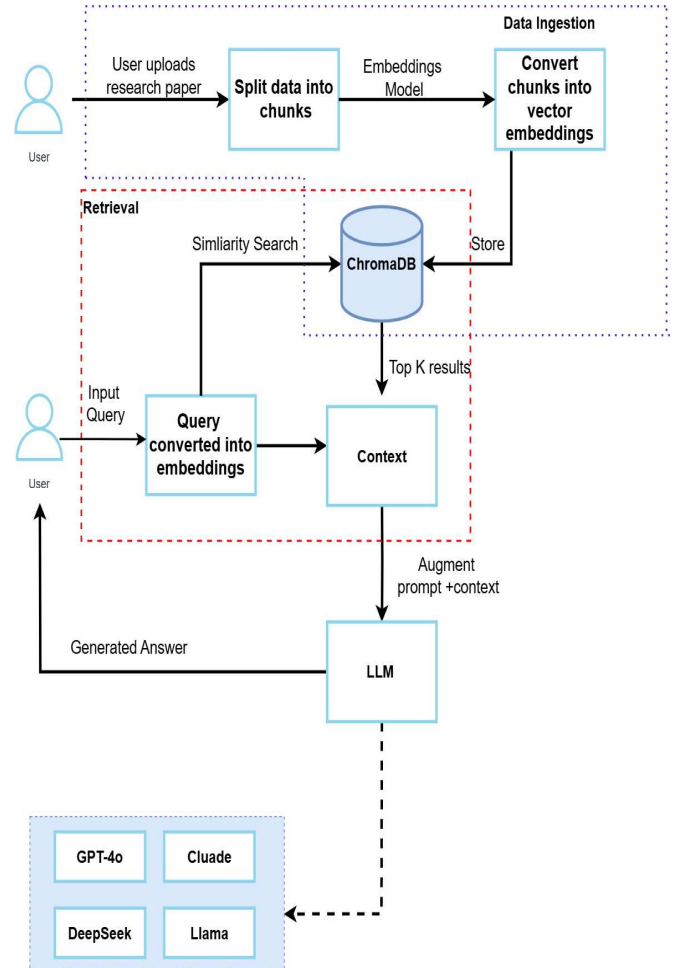


Fig. 1. Flow Chart of the Proposed RAG chatbots

2) *Retriever Phase*: In this phase, the user submits a query relating to the uploaded corpus through the chatbot interface. The information retrieval model converts the query into an embedding and performs a similarity search within the knowledge base to identify relevant and similar embeddings. The retrieved results are then returned as context for further processing. The user is also provided with configurable parameters to refine the retrieval process. The first parameter allows the user to select the top-k results, where k represents the number of highest-ranked results to be returned. The second option relates to the similarity threshold, which refers to the minimum similarity score to be included in the results.

A high threshold reflects more precise and limited context retrieval specific to the given query. The value of k in our systems is five, and that of the similarity level is 20%. This particular step enhances the capability of the system to adjust to user needs by allowing the user to customize the retrieval process according to their needs.

3) *Prompt Augmentation and LLM Integration*: Next, the RAG model augments the user input (or prompts) by adding the relevant retrieved data in context. This step uses prompt engineering techniques to communicate effectively with the LLM. We use a specialized system prompt to help increase the quality of our answers. The augmented prompt allows the large language models to generate an accurate answers to user queries. Embedding the retrieved context into the input prompt allows the LLMs to deliver precise and contextually appropriate answers to user inquiries. We use multiple LLMs, including OpenAI, Claude, Llama, and DeepSeek, for this part of the process. Once the LLM processes the augmented prompt, the resulting output is presented to the user.

B. RAG Models

As mentioned, we used four different LLMs or foundation models to design our RAG system. Below are the complete details of each RAG model. Table I shows a quick overview of the specifics of each RAG chatbot.

- GPT-4o RAG model: OpenAI is one of the most powerful and widely used foundation models. It is currently used to power ChatGPT, a popular AI service. We use OpenAI's GPT-4o model, a versatile, high-intelligence flagship model. It accepts both text and image inputs and produces text outputs (including Structured Outputs). We use this model through API calls validated through a secret API key. For the embedding model, we use OpenAI's text-embedding-ada-002 embeddings.
- Claude RAG model - Anthropic is another front-runner in the foundation models world. We use Anthropic's Claude-3-5-sonnet model, currently the most powerful model in the Claude family. For uniformity, we use OpenAI's text-embedding-ada-002 embeddings model here as well. Claude is also accessed through API calls.
- DeepSeek RAG: DeepSeek R1 is an open-source model. The DeepSeek models are recent additions, and they are from the current year, 2025. Deepseek uses a Mixture-of-Experts (MoE) approach, contrasting the conventionally used transformer-based approaches in models like OpenAI cite [10]. We use the DeepSeek model with 7B parameters. We use the mxbai-embed-large model for the embedding model, which is a highly efficient open-source embedding model. DeepSeek is accessed through Ollama, a framework allowing open-source models to be run locally.
- Llama RAG model: We use the widely popular open-sourced model from the Llama family for our second open-source model and the fourth model in our comparison. Llama models are a family of models made available by MetaAI. We use the Llama 3.1 8B parameter

model locally. We access this model locally through Ollama. For uniformity, we use the mxbai-embed-large model.

TABLE I
SPECIFICATIONS OF RAG-BASED CHATBOTS

RAG System	Base LLM	Embedding Model	Source	Accessed Via
GPT-4o	GPT-4o	OpenAI text-embedding-ada-002	Closed	API
Claude	Claude 3-5 Sonnet	OpenAI text-embedding-ada-002	Closed	API
DeepSeek	DeepSeek R1:7B	mxbai-embed-large	Open	Locally
Llama	Llama 3.1:8B	mxbai-embed-large	Open	Locally

C. Testing Corpus

For evaluation, a collection of research papers from the free and open-access repository arXiv was chosen to check our chatbot system's adaptability and how well it understands academic documents. The papers include many computer science topics like machine learning, networking, cloud computing, and artificial intelligence. This range ensures that the assessment shows the system's capacity to manage many complicated and detailed topics across domains. For this work, we picked ten papers, each presenting complex ideas that demand a detailed understanding, to test the capabilities of the RAG chatbots carefully. The papers are also put into sets of two, resulting in five such groups. Each paper in a set is connected to the other papers by a shared idea or domain. The papers are uploaded in the chatbots in sets of two. Table II shows the domains of papers in each group. The rationale behind this is to have the chatbots answer the questions that have answers that require processing and understanding of both papers.

D. Evaluation Dataset

The crux of our evaluation framework is the uniquely formulated evaluation dataset. Our goal was to design a set that contained questions of varying difficulty that a user could ask the chatbot. The assessment data has 75 questions covering many types, including factual inquiries, yes/no queries, summarization tasks, contrast, and reasoning-based challenges. Usually, understanding literature requires some prior knowledge of other domains or some other literature as well. Typical LLMs seem to give very generic answers when asked to compare and contrast two ideas presented in different

TABLE II
GROUP-WISE DISTRIBUTION OF PAPER DOMAINS

Groups	Paper Domain
1	Information Retrieval
2	Networking and Internet Architecture
3	Machine Learning; Distributed, Parallel, and Cluster Computing
4	Computation and Language
5	Social and Information Networks; Cryptography and Security ; Machine Learning

scientific materials. Therefore, to specifically also test how our RAG systems perform against these complex questions, we also include questions that require an in-depth understanding of multiple research papers in our evaluation dataset. The goal is to understand how well each of our RAG chatbots can understand and reason with the embeddings received from multiple papers. Each group from the set of five has 15 questions designed based on the contents of both papers in the group. Table III shows the total distribution of questions across the evaluation dataset. It highlights a good mix of different types of questions. Fig 2 further shows a detailed breakdown of how these question types are distributed among each group of papers. The system is expected to extract and integrate relevant context from multiple sources for the more complex questions to generate coherent and well-structured answers.

TABLE III
TYPES OF QUESTIONS AND THEIR FREQUENCIES

Type of Questions	No of Questions
Factual Questions (Direct Retrieval from Text)	22
Yes/No Questions (Binary with Justification)	22
Summarization Questions (Condensed Understanding)	8
Reasoning Questions (Complex Analytical Thinking)	11
Contrast Questions (Comparative Analysis)	12

IV. EXPERIMENTS AND RESULTS

The tests were carried out across five separate iterations. In each iteration, we provided all chatbots with two research papers, and all four systems received 15 questions related to each paper pair. There were five rounds, each with 15 queries, across ten documents for each setup. The responses generated by the systems were contrasted with a set of predefined expected responses developed from advice and experience offered by industry practitioners and academics. Following each run, we calculated five important measures to judge system performance.

A. Evaluation Metrics

For this research, we have used the following key metrics.

Question Type Coverage Across Groups

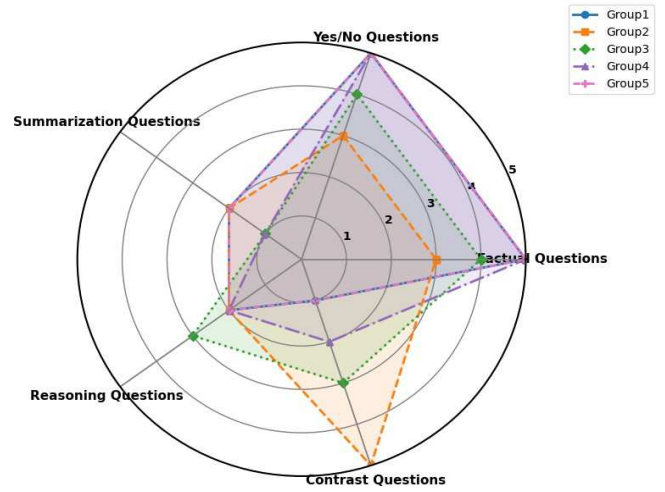


Fig. 2. Distribution of Question Types per Group

- **ROUGE Scores:** ROUGE score is a recall-based metric designed to assess text summarization tasks, where it compares the machine-generated summaries with human-written reference summaries. ROUGE searches for overlapping n-grams, word sequences that appear within both summaries. Unigrams, bigrams, and trigrams are the most commonly used n-grams. Scores range from 0 to 1, with 1 indicating a perfect summary and 0 indicating an incorrect one. ROUGE-N and ROUGE-L are different metrics assessing different aspects of a summary. ROUGE N looks for similar occurrences of n-grams, continuous sequences of n words, with the flexibility of varying the value of n based on the task. This is useful in identifying the grammatical correctness of the summary. ROUGE-L searches for the Longest Common Subsequence between the machine summary and the reference summary. This helps with assessing the quality of the content in the summary. It is the standard in summarization research for its flexibility, objectivity, and correlation with human judgments. For this paper, we use ROUGE-N (ROUGE-1 and ROUGE-2) and ROUGE-L scores.
- **BLEU Scores:** The BLEU score is a precision-based metric that measures overlapping phrases between a machine-translated text and a reference text. It counts the matches of different combinations or phrases of words that occur in both texts. Studies have, however, shown that when used correctly, BLEU provides a good correlation with human judgments of similarities. BLEU scores have a range of 0 to 1, where 1 implies higher similarities and 0 indicates higher differences. In this study, we use BLEU-1 for checking single-word matches and BLEU-4 for testing four-word sequences to capture similarities at the phrase level.
- **BERT Scores:** BERT Scores compare the similarity

between two sentences by calculating the sum of the cosine similarities between their token embeddings. A higher cosine value means a higher BERT score and higher similarities between the texts. It uses BERT, a transformer-based model created by Google, to understand the semantic meaning of the words. It is trained to recognize order and distant parallels of words, thus aiding in the detection of phrases that have the same meaning. BERT Score can handle different types of texts, including pairs, lists, and even single sentences. It combines both precision and recall to provide a precise and balanced measurement of similarities between two texts. BERT has emerged to be an important metric in the field of NLP as it can evaluate the texts based on the meaning of the text, irrespective of the order and words used.

B. Results

Table IV shows the average results of all the metrics obtained using the evaluation dataset against each model in five runs. For more detailed understanding, three graphs corresponding to each metric plotted over the average results are also drawn. Fig 3 shows the comparison of ROUGE-1, ROUGE-2, and ROUGE-L scores of the model. Fig 4 shows the results of the BLEU-1 and BLEU-4 metrics, and Fig 5 shows a line graph plotting the cumulative BERT scores.

TABLE IV
AVERAGE EVALUATION RESULTS OF FIVE TEST RUNS ACROSS VARIOUS METRICS

Metrics	GPT-4o	Claude	DeepSeek	Llama
ROUGE-1	0.44	0.49	0.44	0.42
ROUGE-2	0.23	0.28	0.22	0.20
ROUGE-L	0.31	0.35	0.30	0.27
BLEU-1	0.21	0.25	0.22	0.20
BLEU-4	0.06	0.08	0.05	0.04
BERT	0.89	0.89	0.88	0.87

In Fig 3, Claude achieves the highest scores across ROUGE-1, ROUGE-2, and ROUGE-L, indicating that its generated responses align more closely with the reference text in terms of both core vocabulary and multi-word sequences. The strong ROUGE-2 score suggests that Claude captures key phrases more reliably than the other models, while the higher ROUGE-L value demonstrates a better match in longer subsequences. With somewhat lower but still competitive scores, GPT-4o comes in second, followed closely by DeepSeek and Llama. These results suggest that Claude tends to preserve more essential

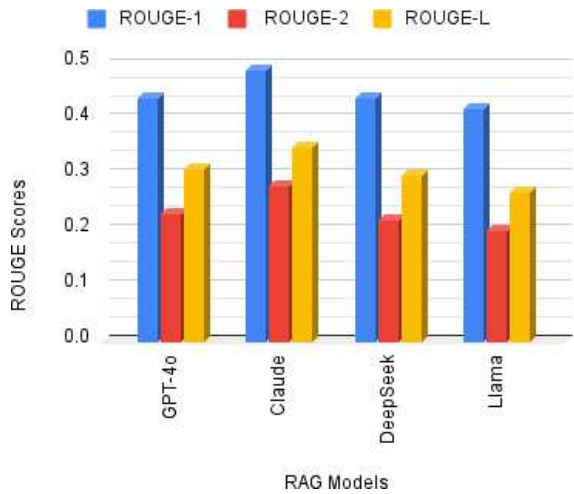


Fig. 3. Average ROUGE-1, ROUGE-2, and ROUGE-L scores of five runs.

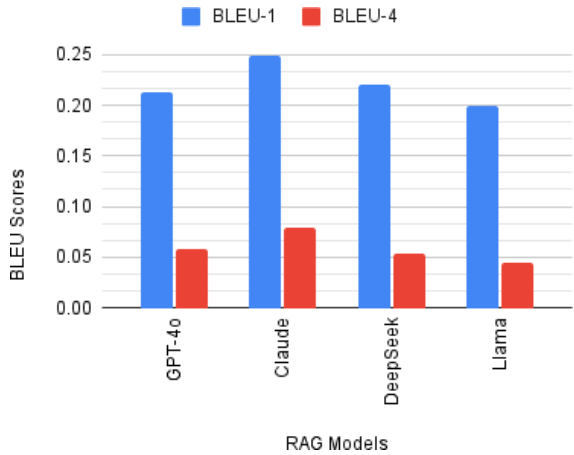


Fig. 4. Average BLEU-1 and BLEU-4 Scores of five runs.

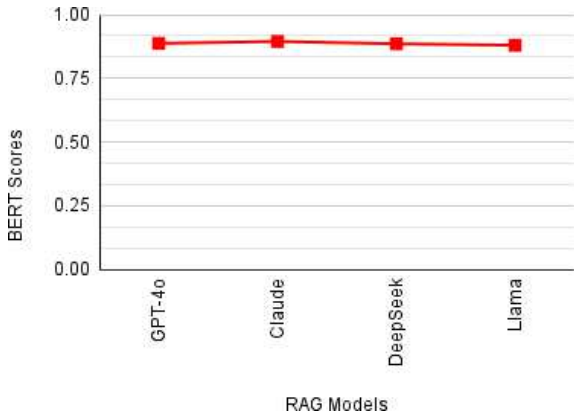


Fig. 5. Average BERT scores of five runs.

words and phrases from the reference answers, highlighting its relative effectiveness among the evaluated models. As shown in Fig 4, BLEU-1 and BLEU-4 scores assessed lexical precision and fluency, respectively. Claude achieved the highest BLEU-1 score, followed by DeepSeek, GPT-4o, and Llama. BLEU-4 scores followed a similar trend, with Claude leading and Llama scoring the lowest. According to these findings, Claude consistently generates responses with more significant lexical overlap, while Llama has trouble with both short- and long-range coherence. Fig 5 shows the BERT scores, which measure semantic similarity by using contextual embeddings rather than depending on candidate summaries. All RAG chatbots had higher BERT scores of 0.85 to 0.90, reflecting strong semantic congruence within the responses. Claude and DeepSeek showed marginally better performance, while Llama showed a marginal reduction. Overall, the Claude RAG chatbot performed better than all the other models combined. GPT-4o's RAG model and DeepSeek performed similarly, with one model outperforming the other in some metrics. Llama performed marginally the worst in all metrics.

V. LIMITATIONS AND FUTURE WORK

This study offers comparative evaluation of RAG chatbots using open-source and close-source language models. However, there are some factors that have possibly influenced the results and are deserving of further investigation. A key factor is a difference in embedding models for open-source chatbots and close-source chatbots that can potentially combine the effects of retrieval and generation. Follow-up research should seek to normalize embeddings to decouple model performance more conclusively. The evaluation data set, while carefully curated by us, cannot possibly encompass the entire variety of user queries or domain complexities encountered in actual chatbot deployments. Its domain distribution, language use, and question difficulty could bias tests in favor of particular strengths of certain models. Expanding the comparison to more representative and diverse datasets like adversarial query sets, multilingual corpora, or open-domain QA benchmarks would strengthen generalization and robust comparative findings. Also, the corpus being academic text in computer science, can be better fit to certain models. For a more balanced study, more materials or sources from various domain can be considered. For further examination, ablation studies that eliminate embedding effects, qualitative error analysis, and user-centric measures such as latency and interpretability are proposed. These areas of investigation would offer a more comprehensive and actionable view of RAG chatbot performance.

VI. CONCLUSION

In this study, we conducted a comparative analysis of four different RAG chatbots based on GPT-4o, Claude Sonnet, DeepSeek, and Llama foundation models. We used a specialized corpus of ten research papers containing ideas from different domains grouped into sets of two. We utilized a specially created evaluation dataset containing different types

of questions to test the range of our chatbots and their ability to generate answers from multiple papers. We then used standard metrics like BLEU, ROUGE, and BERT to compare the responses. Our research concluded that Anthropic's Claude Sonnet model performed the best overall in all metrics, followed by GPT-4o and DeepSeek, and Llama came in last. Among the open-sourced models, DeepSeek performed better than Llama. While models like Claude and GPT-4o perform better, they also come at the cost of transparency and require paid API access. Models like DeepSeek and Llama are open-source, allowing users to access them locally at no cost and providing compelling performance even with fewer parameters than standard closed-source models.

As more powerful LLMs become readily available in the market, studies must be conducted to evaluate the models' effectiveness across a standard benchmark and different domains. Our study hopes to pave the way for further such comparisons, leading to the creation of highly developed RAG systems. Much promise is left to be exploited by these systems, but latency, retrieval bias, and response hallucinations remain open research challenges. Improvements in hybrid retrieval mechanisms, multimodal document processing, and adaptive learning algorithms will further entrench the role of RAG chatbots in the academic world.

REFERENCES

- [1] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, 2020, pp. 9459–9474.
- [2] B. Tural, Z. O' rpek, and Z. Destan, "Retrieval-Augmented Generation (RAG) and LLM Integration," in *2024 8th International Symposium on Innovative Approaches in Smart Technologies (ISAS)*, Istanbul, Turkey: IEEE, Dec. 2024, pp. 1–5. doi: 10.1109/ISAS64331.2024.10845308.
- [3] M. Dean, R. R. Bond, M. F. McTear, and M. D. Mulvenna, "ChatPapers: An AI chatbot for interacting with academic research," in *2023 31st Irish Conference on Artificial Intelligence and Cognitive Science (AICS)*, IEEE, Dec. 2023, pp. 1–7.
- [4] M. Maryamah, M. M. Irfani, E. B. Tri Raharjo, N. A. Rahmi, M. Ghani, and I. K. Raharjana, "Chatbots in academia: A retrieval-augmented generation approach for improved efficient information access," in *2024 16th International Conference on Knowledge and Smart Technology (KST)*, IEEE, Feb. 2024, pp. 259–264.
- [5] U. S. Ali, "Ask your transcript: LLM-driven insights for academic advising," in *2024 2nd International Conference on Computing and Data Analytics (ICCCA)*, IEEE, Nov. 2024, pp. 1–4.
- [6] D. Joshi, S. Tekade, S. Zanzane, D. Sakharwade, A. Tripathi, and S. Tiwadi, "Generative AI-driven chatbot for personalized university information and assistance," in *2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAQSA)*, IEEE, Dec. 2024, pp. 1–6.
- [7] R. Shan, "OpenRAG: Open-source retrieval-augmented generation architecture for personalized learning," in *2024 4th International Conference on Artificial Intelligence, Robotics, and Communication (ICAIRC)*, IEEE, Dec. 2024, pp. 212–216.
- [8] M. A. Khadija, A. Aziz, and W. Nurharjadmo, "Automating information retrieval from faculty guidelines: Designing a PDF-driven chatbot powered by OpenAI ChatGPT," in *2023 International Conference on Computer, Control, Informatics and its Applications (IC3INA)*, IEEE, Oct. 2023, pp. 394–399.
- [9] "Chroma: the AI-native open-source embedding database." Github.
- [10] DeepSeek-AI et al., "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning," Jan. 22, 2025, arXiv: arXiv:2501.12948. doi: 10.48550/arXiv.2501.12948.